

Qualifying Exam in Microeconomics
Arizona State University
June 2019

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. Consider the following two person game Γ where the subindex on a strategy label indicates the player.

	a_2	b_2	c_2
a_1	(5, 3)	(0, 0)	(2, 0)
b_1	(0, 0)	(2, 2)	(0, 0)
c_1	(0, 0)	(0, 0)	(0, 0)

- (a) Identify all pure strategy Nash equilibria of Γ .
 - (b) Let $\Gamma(2)$ be the game Γ repeated twice. Player i 's payoffs are the average of the two repetitions. After each instance of Γ , both players observe the outcome. Identify the pure strategy Subgame Perfect Equilibrium (SPE) outcomes of $\Gamma(2)$ and show SPE strategies that support them.
 - (c) Is there an SPE outcome of $\Gamma(3)$ (i.e. Γ repeated three times) where player 2's average payoff is 1.5? Is there an SPE outcome of $\Gamma(4)$ (i.e. Γ repeated four times) where player 2's average payoff is 1.5?
2. The two only firms in a given industry must decide their respective advertising expenditures. Profit for firm i , $i = 1, 2$, is

$$\pi_i(x_i, x_j) = x_i(2 - x_i) - x_j,$$

where x_i and x_j are, respectively, firm i 's and j 's advertising expenditures.

- (a) Suppose both firms choose their expenditures simultaneously and independently. Please find a Nash equilibrium in pure strategies. What are the Nash equilibrium profits?
- (b) Is the solution found in (a) Pareto Optimal? Please compute the Pareto optimal symmetric solution (i.e., both firms obtain the same profit). Why does the solution to (a) fail to be Pareto optimal?
- (c) Suppose now that firms do not act independently. Each firm can guarantee for itself a profit of zero by withdrawing from the market (i.e., the reservation profit is zero). Suppose the firms will only agree on an outcome that is on the core. Please characterize (i.e., describe mathematically) the solutions that are in the core.

3. Consider the following principal-agent problem with moral hazard, where the principal is risk neutral and the agent strictly risk averse. The agent's utility of income u is given by $u(w) = 2\sqrt{w}$, $w \geq 0$, and his disutility of effort ψ is given by $\psi(a) = a^2/2$, with $a \in [0, 0.4]$, so that if he exerts effort a and obtains wage w then his utility is $2\sqrt{w} - 0.5a^2$. For each effort level a , the output x is distributed on $[0, 1]$ according to $f(\cdot|a)$ given by $f(x|a) = 1 + 0.5(1 - 2x)(1 - 2a)$. Finally, the agent's reservation utility is denoted by $\bar{u} > 0$ (assumed to be large enough so that the participation constraint can be guaranteed to bind at the optimum and wages are positive).

- (a) Set up the principal's optimal contracting problem.
- (b) Does the monotone likelihood ratio property (*MLRP*) hold? Is the first-order approach (*FOA*) valid? Justify your answers formally.
- (c) Solve the principal's cost-minimization problem for each a , obtain the cost-minimizing contract (in wage or utility terms). Is it increasing in x ? Derive the cost function C .
(Hint: At some point you may want to use $\int_0^1 \frac{(1-2x)}{1+0.5(1-2x)(1-2a)} dx = \frac{a+0.5\left(\log\left(\frac{0.75-0.5a}{0.25+0.5a}\right)-1\right)}{(0.5-a)^3}$.)
- (d) Set up the principal's profit maximization problem. Is the optimal a characterized by the Kuhn-Tucker conditions? (If the algebra is too messy simply explain clearly how you would proceed to answer this part of the problem.)
- (e) Suppose that the principal could observe, in addition to x , a signal $y \in [0, 1]$ distributed according with a density g and cumulative distribution G . Assume that, for each a , the joint density of (x, y) is given by

$$q(x, y|a) = f(x|a)g(y) (1 + \alpha(1 - 2F(x|a))(1 - 2G(y))),$$

where $\alpha \in (-1, 1)$. Will the principal include y in the agent's compensation? Explain.

4. Consider the following competitive screening model of a credit market. There are two identical lenders who offer loan contracts. A loan contract consists of a pair (r, x) , $r \geq 0$ and $x \in [0, 1]$, where r is the interest on a loan of size one given to the borrower to finance a project that has value one, and x is the amount of collateral that the borrower is asked to deposit when he receives the loan. Each borrower has a probability q of becoming insolvent (q is the borrower's credit risk) and hence unable to repay back the loan (principal or interest). In that event the lender seizes the collateral x . There is, however, an inefficiency in realizing the collateral: to a lender, a collateral x only has value bx , $0 < b < 1$. In short, the payoff for a borrower if he accepts contract (r, x) is $1 - r(1 - q) - qx$ and for the lender is $r(1 - q) - qbx$. There is a fraction $\lambda \in (0, 1)$ of borrowers who are of low credit risk q_ℓ and $1 - \lambda$ are of high credit risk q_h , with $0 < q_\ell < q_h < 0.5$. The probability q_i , $i = \ell, h$, is private information of the borrower. Both lenders and borrowers are risk neutral and the borrowers' reservation utility is zero. The timing is as follows: lenders offer any finite number of contracts; borrowers observe the contracts and take the one that suits them best (or reject all of them), with ties

broken equiprobably; nature reveals whether a borrower is insolvent or not; payoffs are realized.

- (a) Assume first that q_i , $i = \ell, h$, is observable. Show that in this case any SPE entails zero collateral and $r_i = q_i/(1 - q_i)$, $i = \ell, h$. Draw a picture.
- (b) Derive the unique SPE candidate menu of contracts $((r_\ell, x_\ell), (r_h, x_h))$. Using pictures (with r on the vertical axis and x on the horizontal axis), explain why this is the unique candidate. Does a SPE exist? Explain.
- (c) Suppose that the two lenders collude and behave like a monopoly, offering a menu $((r_\ell^m, x_\ell^m), (r_h^m, x_h^m))$ to maximize profits and then divide them equally between the two of them. Write down (without solving) the problem that these firms would solve as a monopolist.